ISE 555: Unit 4 Assignment

Fall 2026

For this assignment, all problems, including the writing assignment (Problem 5), are mandatory to receive full credit. A PDF file containing all your solutions as well as the writing assignment must be submitted on Brightspace by **4:00 pm on Monday, October 12th.** Make sure you have access to your submission during class on the day that the assignment is due. This will be helpful when presenting your solutions to the class in person.
Please type your solutions if possible. Suggest to use LaTeX (available https://overleaf.com)

1) Solve the following problem using the Simplex Method. Show your tableau at each iteration.

$$\begin{aligned} \text{minimize} \quad & z = -5x_1 - 7x_2 - 12x_3 + x_4 \\ \text{subject to} \quad & 2x_1 + 3x_2 + 2x_3 + x_4 \leq 38 \\ & 3x_1 + 2x_2 + 4x_3 - x_4 \leq 55 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

2) Solve the following problem using the Simplex Method. Show your tableau at each iteration.

$$\begin{aligned} \text{maximize} \quad & z = 5x_1 + 3x_2 + 2x_3 \\ \text{subject to} \quad & 4x_1 + 5x_2 + 2x_3 + x_4 \leq 20 \\ & 3x_1 + 4x_2 - x_3 + x_4 \leq 30 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

3) Solve the following problem using the Simplex Method. Show your tableau at each iteration.

$$\begin{aligned} \text{maximize} \quad & z = 3x_1 + 9x_2 \\ \text{subject to} \quad & -5x_1 + 2x_2 \leq 30 \\ & -3x_1 + x_2 \leq 12 \\ & x_1, x_2 \geq 0 \end{aligned}$$

4) Solve the following problem using the Simplex Method. Show your tableau at each iteration.

$$\begin{aligned} \text{maximize} \quad & z = 7x_1 + 8x_2 \\ \text{subject to} \quad & 4x_1 + x_2 \leq 100 \\ & x_1 + x_2 \leq 80 \\ & x_1 \leq 40 \\ & x_1, x_2 \geq 0 \end{aligned}$$

5) WRITING ASSIGNMENT: Your company's management team has finally bought in to using optimization as a way to improve production. Write a paragraph explaining at least one caution to them about why optimization may not always be helpful (i.e. explain an issue that may cause optimization to give a non-optimal result).